OLD MUTUAL
SOUTH AFRICAN MATHEMATICS OLYMPIAD

Organised by the
SOUTH AFRICAN MATHEMATICS FOUNDATION

2023 FIRST ROUND
SENIOR SECTION: GRADE 10 - 12

9 March 2023 **Time: 60 minutes** **Number of questions: 20**

Instructions

1. *This is a multiple choice question paper. Each question is followed by five answers marked A, B, C, D and E. Only one of these is correct.*
2. *Scoring rules:*
   2.1. *Each correct answer is worth 5 marks.*
   2.2. *There is no penalty for an incorrect answer or any unanswered question.*
3. *You must use an HB pencil. Rough work paper, a ruler and an eraser are permitted.* ***Calculators and geometry instruments are not permitted.***
4. *Figures are not necessarily drawn to scale.*
5. *Indicate your answers on the sheet provided.*
6. *Start when the invigilator tells you to do so.*
7. *Answers and solutions will be available at* `www.samf.ac.za`

Do not turn the page until you are told to do so.
Draai die boekie om vir die Afrikaanse vraestel.

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Organisations involved: AMESA, SA Mathematical Society,
SA Akademie vir Wetenskap en Kuns, ASTEMI

The Mathematics Talent Search is a free online problem-solving course for learners from Gr 7 – 12 presented by the SAMF. All you have to do to participate is to click on https://mytutor.chat/samf-talent-search/ or to scan the QR code to go to the MyTutor.chat site easily.

1. How many zeros are there in $\dfrac{10^{2023}}{10^{2022}}$ when written in simplest form?

   (A) 1 (B) 2 (C) 3 (D) 2022 (E) 2023

2. Which of the following three statements are true?

   (i) All prime numbers are odd.

   (ii) All cubes of integers are odd.

   (iii) There is an infinite number of positive even integers.

   (A) only (i) (B) only (ii) (C) (i) and (ii) (D) only (iii) (E) (i) and (iii)

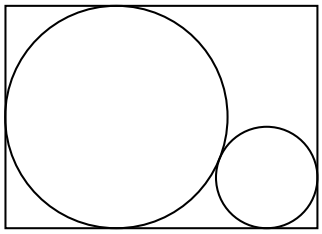
3. A plane takes 8 hours and 20 minutes to fly from Durban to Dubai. Dubai's local time is 2 hours ahead of Durban's local time (e.g., when it is 12:00 in Durban it is already 14:00 in Dubai). If the plane departs from Durban at 09:55, find its time of arrival in Dubai.

   (A) 20:15 (B)19:15 (C) 18:15 (D) 17:15 (E) 16:15

4. In $\triangle PQR$ the angle at P is three times the angle at Q and half the angle at R. What is the angle at P in degrees?

   (A) 30 (B) 36 (C) 54 (D) 60 (E) 72

5. Two circular pieces of wood are put inside a rectangular framework with side lengths 25 and 18. The circles touch each other as well as the sides of the framework as shown in the diagram. What is the area of the largest wooden circle?



   (A) 18π (B) 81π (C) 25π (D) 49π (E) 7π

6. Sarah starts typing 60 words per minute on her computer. Her fingers get tired after 3 minutes but, nevertheless, she continues typing; now at a rate 25% slower than her original rate. How long, in minutes, will it take her to type 900 words?

   (A) 19 (B) 16 (C) 27 (D) 54 (E) 144

7. If $17x + 51y = 85$, what is the value of $19x + 57y$?

   (A) 5 (B) 95 (C) 10 (D) 85 (E) 93

8. The average (mean) of $\frac{1}{5}$ and $\frac{1}{10}$ is $\frac{1}{x}$. The value of x is

   (A) $\frac{2}{15}$ (B) $\frac{10}{3}$ (C) 30 (D) $\frac{1}{7}$ (E) $\frac{20}{3}$

9. Two waitresses, Debbie and Refilwe, would like to go to the movies together on a night when they are both off work. Debbie has every eighth night off counting from today, while Refilwe has every sixth night off counting from tomorrow. In how many days from today can they go to the movies together?

   (A) 1 (B) 2 (C) 24 (D) 48 (E) Never

10. The smallest integer by which you must multiply 2023 to get a perfect square is

    (A) 23 (B) 17 (C) 7 (D) 2023 (E) 119

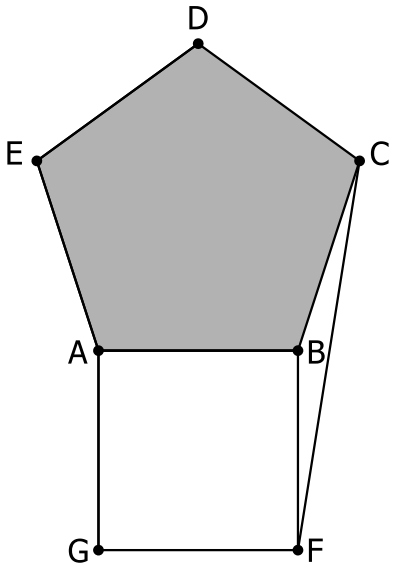
11. On Saturday Hippy went to Carnival City to gamble. He started with R20 and doubled his money every hour. He gambled for 20 hours before going home with his fortune. On the following day, Sunday, Happy went to Carnival City. He started gambling with R80 and doubled his money every hour. He gambled until his fortune matched Hippy's. For how many hours did Happy gamble?

    (A) 10 (B) 17 (C) 19 (D) 18 (E) 5

12. Each of the 12 edges of a cube is coloured either red or green. Every face of the cube has at least one red edge. What is the smallest possible number of red edges?

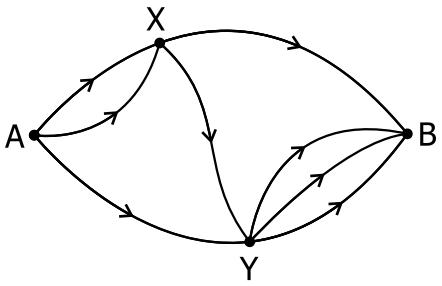
    (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

13. A regular pentagon and a square share a common side, AB, as shown in the diagram. What is the size of $\angle BFC$ in degrees?



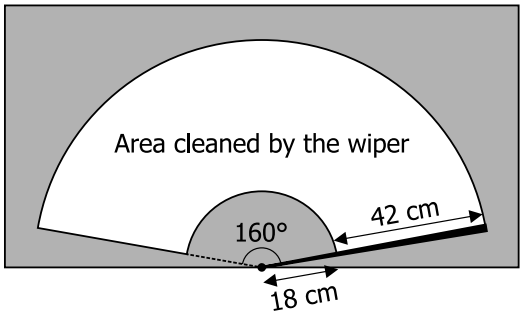
(A) 15 (B) 11 (C) 7 (D) 9 (E) 12

14. In the diagram, it is possible to travel along a path, only in the direction indicated by the arrows. Jannecke studied the figure, and determined all the possible routes from A to B. She selected one of these routes at random. What is the probability that she selected a route which passes through X?



(A) $\frac{8}{11}$ (B) $\frac{3}{11}$ (C) 1 (D) $\frac{9}{11}$ (E) $\frac{6}{11}$

15. The diagram models a rectangular windscreen of a car. The windscreen wiper can rotate through 160°. What is the area, in square cm, cleaned by the wiper?

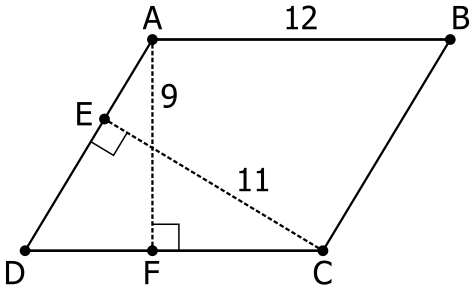


(A) 144π (B) 484π (C) 1312π (D) 1456π (E) 1600π

16. If a, b and c are positive integers, find the area (in square units) of the triangle whose vertices are $(a, b+c)$, $(a, b-c)$ and $(-a, c)$.

(A) $2ac$ (B) $2bc$ (C) $b(a+c)$ (D) $c(a-b)$ (E) $2+2b+c$

17. The length of AB in parallelogram ABCD is 12. The perpendicular lines to the sides CD and AD are respectively 9 and 11. Find the length of AD.

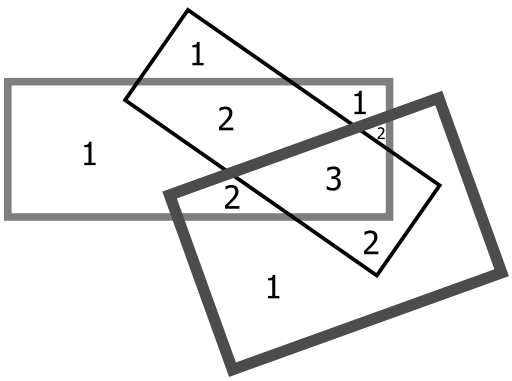


(A) $\frac{108}{11}$ (B) $\frac{108}{10}$ (C) $\frac{99}{10}$ (D) 10 (E) $\frac{99}{12}$

18. A straight one-way city street has 8 consecutive traffic lights. Every light remains green for 1.5 minutes, yellow for 3 seconds, and red for 1.5 minutes. The lights are synchronised so that each light turns red 10 seconds after the preceding one turns red. What is the longest interval of time, in seconds, during which all 8 lights are green?

(A) 10 (B) 15 (C) 20 (D) 25 (E) 30

19. William delivers three rugs at a school. He puts them on the floor in the school hall in such a way that they partially overlap (1-layer parts, 2-layer parts and 3-layer parts) as indicated in the figure. Now an area of 64 m^2 is covered. The individual areas of the three rugs added together, is 150 m^2. The area covered by exactly three layers is 28 m^2. What is the area covered by exactly two layers (in m^2)?



(A) 30 (B) 36 (C) 56 (D) 60 (E) 64

20. A terrible dragon has three heads and three tails. A knight must kill it by cutting off all its heads and tails with his sword. With one stroke of his sword, he can cut off either one or two heads, or either one or two tails. When he cuts off one head, a new one grows in its place. When he cuts off two heads, nothing grows. When he cuts off one tail, two new tails grow. When he cuts off two tails, one new head grows. What is the least number of strokes with the sword needed to kill the dragon?

(A) 3 (B) 5 (C) 7 (D) 9 (E) 11

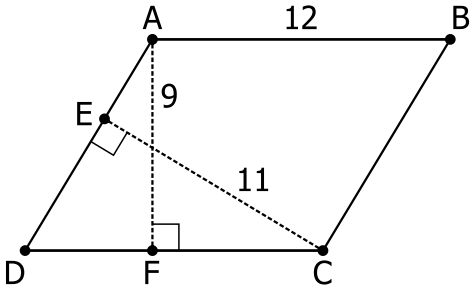
20. 'n Vreeslike draak het drie koppe en drie sterte. 'n Ridder moet die draak doodmaak deur al sy koppe en sterte met 'n swaard af te kap. Met een hou van sy swaard kan die ridder óf een óf twee koppe afkap, óf een óf twee sterte. As hy een kop afkap, groei 'n nuwe kop in sy plek. As hy twee koppe afkap, groei niks daarna nie. As hy een stert afkap, groei twee nuwe sterte. As hy twee sterte afkap, groei een nuwe kop. Wat is die kleinste hoeveelheid kappe wat die ridder met sy swaard moet maak om die draak dood te maak?

(A) 3 (B) 5 (C) 7 (D) 9 (E) 11

16. As a, b en c positiewe heelgetalle is, vind die oppervlakte (in vierkante eenhede) van die driehoek met hoekpunte $(a, b+c)$, $(a, b-c)$ en $(-a, c)$.

(A) $2ac$ (B) $2bc$ (C) $b(a+c)$ (D) $c(a-b)$ (E) $2+2b+c$

17. Die lengte van AB in parallelogram $ABCD$ is 12. Die loodregte lyne op die sye CD en AD is onderskeidelik 9 en 11. Vind die lengte van AD.

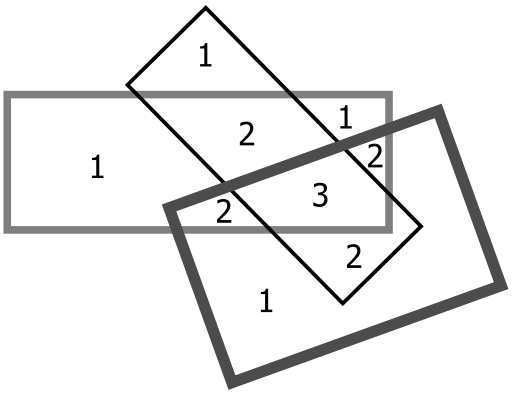


(A) $\frac{108}{11}$ (B) $\frac{108}{10}$ (C) $\frac{99}{10}$ (D) 10 (E) $\frac{99}{12}$

18. 'n Reguit eenrigtingstraat in 'n stad het 8 opeenvolgende verkeersligte. Elke lig bly vir 1.5 minute groen, is geel vir 3 sekondes en rooi vir 1.5 minute. Die ligte is gesinchroniseer sodat elke lig rooi word 10 sekondes nadat die vorige lig rooi geword het. Wat is die langste aaneenlopende tyd, in sekondes, waartydens al 8 ligte groen was?

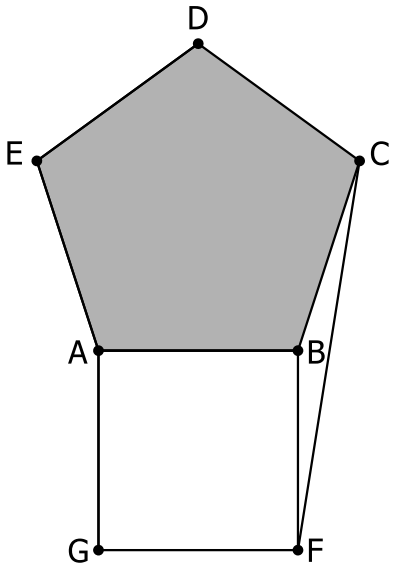
(A) 10 (B) 15 (C) 20 (D) 25 (E) 30

19. William het drie matte by 'n skool afgelewer. Hy het hulle só op die vloer neergesit dat hulle gedeeltelik oorvleuel het (daar is dele met een laag, dele met twee lae en dele met drie lae) soos aangedui in die figuur. Nou is 'n oppervlakte van 64 m^2 bedek. Die individuele oppervlaktes van die drie matte saam is 150 m^2. Die oppervlakte wat deur presies drie lae bedek is, is 28 m^2. Wat is die oppervlakte (in m^2) wat deur presies twee lae bedek word?



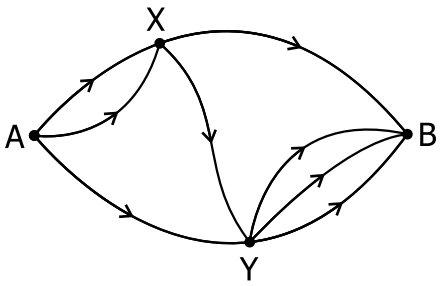
(A) 30 (B) 36 (C) 56 (D) 60 (E) 64

13. 'n Reëlmatige vyfhoek en 'n vierkant deel 'n gemeenskaplike sy, AB, soos aangetoon in die diagram. Wat is die grootte van $\angle BFC$ in grade?



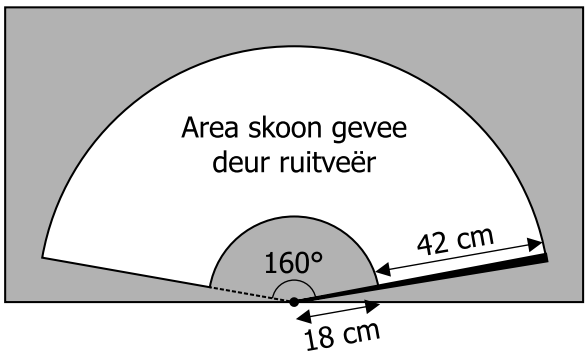
(A) 15 (B) 11 (C) 7 (D) 9 (E) 12

14. In die diagram kan 'n mens op elke pad slegs in die rigting reis waarheen die pyltjies wys. Jannecke het die diagram bestudeer en al die moontlike roetes van A na B bepaal. Sy het een van die roetes ewekansig gekies. Wat is die waarskynlikheid dat die roete wat sy gekies het deur X gaan?



(A) $\frac{8}{11}$ (B) $\frac{3}{11}$ (C) 1 (D) $\frac{9}{11}$ (E) $\frac{6}{11}$

15. Die diagram stel 'n reghoekige windskerm (voorruit) van 'n motor voor. Die ruitveër van die windskerm kan deur 160° roteer. Wat is die oppervlakte, in vierkante cm, wat deur die ruitveër skoongevee word?



(A) 144π (B) 484π (C) 1312π (D) 1456π (E) 1600π

7. As $17x + 51y = 85$, wat is die waarde van $19x + 57y$?

   (A) 5 (B) 95 (C) 10 (D) 85 (E) 93

8. Die gemiddelde van $\frac{1}{5}$ en $\frac{1}{10}$ is $\frac{1}{x}$. Die waarde van x is

   (A) $\frac{2}{15}$ (B) $\frac{10}{3}$ (C) 30 (D) $\frac{1}{7}$ (E) $\frac{20}{3}$

9. Twee kelnerinne, Debbie en Refilwe, wil graag een aand gaan fliek wanneer hulle saam af het by die werk. Debbie het elke agtste aand af, getel vanaf vandag, terwyl Refilwe elke sesde aand af het, getel vanaf môre. Na hoeveel dae vanaf vandag kan hulle saam gaan fliek?

   (A) 1 (B) 2 (C) 24 (D) 48 (E) Nooit

10. Die kleinste heelgetal waarmee jy 2023 moet vermenigvuldig om 'n volkome vierkant te kry, is

    (A) 23 (B) 17 (C) 7 (D) 2023 (E) 119

11. Saterdag gaan Hippy na Carnival City om te dobbel. Hy begin met R20 en verdubbel sy geld elke uur. Hy dobbel vir 20 uur voor hy huis toe gaan. Die volgende dag, Sondag, gaan Happy na Carnival City. Hy begin met R80 en verdubbel sy geld elke uur. Hy dobbel totdat sy fortuin gelyk is aan Hippy s'n. Hoeveel uur het Happy gedobbel?

    (A) 10 (B) 17 (C) 19 (D) 18 (E) 5

12. Elk van die 12 kante (edges) van 'n kubus is óf rooi óf groen ingekleur. Elke vlak van die kubus het ten minste een rooi kant. Wat is die kleinste moontlike getal rooi kante?

    (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

1. Hoeveel nulle is daar in $\dfrac{10^{2023}}{10^{2022}}$ as dit in eenvoudigste vorm uitgedruk word?

   (A) 1 (B) 2 (C) 3 (D) 2022 (E) 2023

2. Watter van die volgende drie stellings is waar?

   (i) Alle priemgetalle is onewe.

   (ii) Alle heelgetalderdemagte is onewe.

   (iii) Daar is 'n oneindige hoeveelheid positiewe ewe heelgetalle.

   (A) slegs (i) (B) slegs (ii) (C) (i) en (ii) (D) slegs (iii) (E) (i) en (iii)

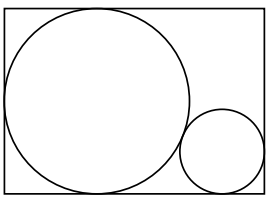
3. Dit neem 'n vliegtuig 8 uur en 20 minute om van Durban na Dubai te vlieg. Die plaaslike tyd in Dubai is 2 uur voor die plaaslike tyd in Durban (bv., wanneer dit 12:00 is in Durban is dit reeds 14:00 in Dubai). Hoe laat kom die vliegtuig in Dubai aan as dit om 09:55 in Durban opgestyg het?

   (A) 20:15 (B) 19:15 (C) 18:15 (D) 17:15 (E) 16:15

4. In $\triangle PQR$ is die hoek by P drie keer die hoek by Q en die helfte van die hoek by R. Hoe groot is die hoek by P in grade?

   (A) 30 (B) 36 (C) 54 (D) 60 (E) 72

5. Twee sirkelvormige stukke hout word in 'n reghoekige raamwerk met sylengtes 25 en 18 gesit. Die sirkels raak mekaar sowel as die raamwerk soos aangetoon in die diagram. Wat is die oppervlakte van die grootste houtsirkel?



   (A) 18π (B) 81π (C) 25π (D) 49π (E) 7π

6. Sarah begin tik teen 60 woorde per minuut op haar rekenaar. Na 3 minute is haar vingers moeg, maar nogtans hou sy aan met tik; nou teen 'n spoed 25% stadiger as haar oorspronklike spoed. Hoe lank sal dit haar neem, in minute, om 900 woorde te tik?

   (A) 19 (B) 16 (C) 27 (D) 54 (E) 144

OLD MUTUAL
SUID-AFRIKAANSE WISKUNDE-OLIMPIADE

Georganiseer deur die
SOUTH AFRICAN MATHEMATICS FOUNDATION

2023 EERSTE RONDTE
SENIOR AFDELING: GRAAD 10-12

9 Maart 2023 **Tyd: 60 minute** **Aantal vrae: 20**

Instruksies

1. *Hierdie is 'n veelvuldige-keuse vraestel. Na elke vraag is vyf antwoorde, genommer A, B, C, D en E. Net een van hulle is reg.*
2. *Puntetoekenning:*
   2.1. *Elke korrekte antwoord tel 5 punte.*
   2.2. *Daar is geen penalisering vir foutiewe antwoorde of vrae wat nie beantwoord is nie.*
3. *Gebruik 'n HB potlood. Papier vir rofwerk, 'n liniaal en uitveër word toegelaat.* ***Sakrekenaars en meetkunde-instrumente word nie toegelaat nie.***
4. *Figure is nie noodwendig volgens skaal geteken nie.*
5. *Beantwoord die vrae op die antwoordblad wat voorsien word.*
6. *Begin sodra die toesighouer die teken gee.*
7. *Antwoorde en oplossings sal beskikbaar wees by* `www.samf.ac.za`

Moenie omblaai voordat dit aan jou gesê word nie.
Turn the booklet over for the English paper.

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Organisasies betrokke: AMESA, SA Wiskundevereniging,
SA Akademie vir Wetenskap en Kuns

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Al wat jy moet doen om deel te neem, is om te klik op https://mytutor.chat/samf-talent-search/ of skandeer die QR kode om maklik na MyTutor.chat te gaan.